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pip install opencv-python matplotlib numpy
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Requirement already satisfied: opencv-python in /usr/local/lib/python3.12/dist-packages (4.13.0.92)
Requirement already satisfied: matplotlib in /usr/local/lib/python3.12/dist-packages (3.10.0)
Requirement already satisfied: numpy in /usr/local/lib/python3.12/dist-packages (2.0.2)
Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (1.3.3)
Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (4.63.0)
Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (1.5.0)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (26.2)
Requirement already satisfied: pillow>=8 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (11.3.0)
Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (3.3.2)
Requirement already satisfied: python-dateutil>=2.7 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (2.9.0.post0)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.7->matplotlib) (1.17)
```

```
import cv2                                # Import OpenCV library for image processing
import matplotlib.pyplot as plt            # Import Matplotlib for displaying images

# Read the image from the disk into a NumPy array
img = cv2.imread('number_plate.jpg')      # cv2.imread reads the image in BGR (Blue-Green-Red) format
# Convert the BGR image to RGB format since Matplotlib expects RGB
img_rgb = cv2.cvtColor(img, cv2.COLOR_BGR2RGB) # This ensures colors look correct when displayed

# Display the image using Matplotlib
plt.imshow(img_rgb)                        # Render the RGB image array
plt.title('Original Image')               # Add a title to the plot window
plt.axis('off')                           # Hide the X and Y axes (ticks and borders) for a clean look
plt.show()                                # Show the actual window on screen

# Print the dimensions of the loaded image
print("Image Shape:", img.shape)           # Outputs (height, width, channels) e.g., (100, 400, 3)
```

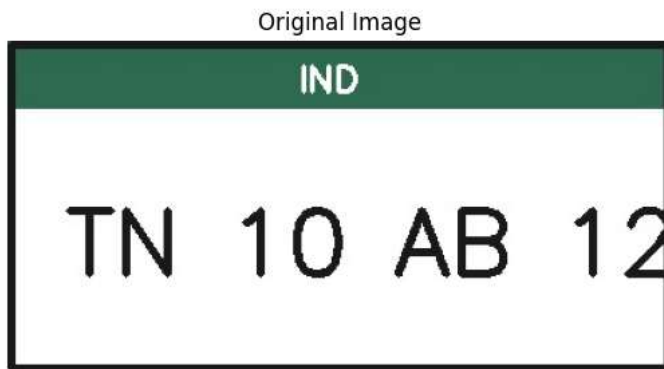


Image Shape: (200, 400, 3)

```
# Downsample the image to a tiny 50x25 resolution to simulate a very low sampling rate (bad camera sensor)
low_res = cv2.resize(img, (50, 25), interpolation=cv2.INTER_NEAREST) # INTER_NEAREST simply drops pixels without blending
print("Low-Res Shape:", low_res.shape)                                # The image is now only 25 pixels high and 50 pixels wide

# Scale the tiny image back up to 400x200 so we can actually see the blocky pixelation effect on screen
pixelated = cv2.resize(low_res, (400, 200), interpolation=cv2.INTER_NEAREST) # Upscale using Nearest Neighbor to keep it blocky

# Create a figure with a specific size (width=10, height=4 inches)
plt.figure(figsize=(10, 4))                                             # Create a visualization window to hold our images

# Create the left subplot (1 row, 2 columns, 1st position)
plt.subplot(1, 2, 1)                                                    # Position this image within a grid layout
plt.imshow(cv2.cvtColor(img, cv2.COLOR_BGR2RGB)) # Display the original, high-quality image
plt.title('Original (High Sampling)')                                   # Set a descriptive title for this plot
plt.axis('off')                                                         # Hide axis ticks and coordinates

# Create the right subplot (1 row, 2 columns, 2nd position)
plt.subplot(1, 2, 2)                                                    # Position this image within a grid layout
plt.imshow(cv2.cvtColor(pixelated, cv2.COLOR_BGR2RGB)) # Display the blurred/blocky low-sampling image
plt.title('Pixelated (Low Sampling)')                                   # Set a descriptive title for this plot
plt.axis('off')                                                         # Hide axis ticks and coordinates
plt.show()                                                              # Render both subplots to the screen
```

Low-Res Shape: (25, 50, 3)

Original (High Sampling)



Pixelated (Low Sampling)



```
# Define a list of different resolutions to test (Low, Medium, High)
sizes = [(50, 25), (200, 100), (400, 200)] # Width x Height tuples
# Labels for our plot titles
labels = ['Low (50x25)', 'Medium (200x100)', 'High (400x200)'] # Define corresponding text labels

plt.figure(figsize=(14, 4)) # Create a wide figure window

# Loop through each size and label simultaneously
for i, (size, label) in enumerate(zip(sizes, labels)): # Iterate through the parameters
    # Step 1: Downsample the original image to the specific 'size' to simulate the sensor resolution
    small = cv2.resize(img, size, interpolation=cv2.INTER_AREA) # INTER_AREA is best for shrinking images

    # Step 2: Upsample it back to 400x200 so we can compare them all at the same physical size
    restored = cv2.resize(small, (400, 200), interpolation=cv2.INTER_NEAREST) # Upsample image

    plt.subplot(1, 3, i + 1) # Create a subplot (1 row, 3 columns, position i+1)
    plt.imshow(cv2.cvtColor(restored, cv2.COLOR_BGR2RGB)) # Show the image
    plt.title(label) # Set the title to the current sampling rate label
    plt.axis('off') # Hide the axis

plt.suptitle('Effect of Different Sampling Rates') # Add a main title for the whole window
plt.tight_layout() # Automatically adjust spacing between plots
plt.show() # Display the final comparison
```

Effect of Different Sampling Rates

Low (50x25)



Medium (200x100)



